Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Mathematics II

BCA — Semester II — 2022 — Answer Sheet

Subject Code: CACS154

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. "A function $f(x)$ is defined as $f(x) = \begin{cases} 2x + 3 & \text{for } -\frac{3}{2} \leq x < 0 \\ 3 - 2x & \text{for } 0 \leq x \leq \frac{3}{2} \\ -3 - 2x & \text{for } x > \frac{3}{2} \end{cases}$ show that $f(x)$ is continuous at $x = 0$ and discontinuous at $x = \frac{3}{2}$."

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Find $\frac{dy}{dx}$ when $x = a(t + \sin t)$, $y = a(1 - \cos t)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. State L'Hospital's Rule. Use it to evaluate: $\lim_{x \to 0} \frac{xe^x - In(1+x)}{x^2}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. State Rolle's Theorem. Verify Rolle's Theorem for the function $f(x) = x^2 - 9$ in the interval $-3 \leq x \leq 3$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Evaluate:a) $\int_0^2 \frac{dx}{x^2-25}$b) $\int \frac{2x+5}{\sqrt{x^2}\ + 5x} dx$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Solve: $\sin x \frac{dy}{dx} + (\cos x) y = -\sin x \cos x$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the following system by Gauss-Seidel method: $\begin{aligned} -4x + y - z &= -8 \\ 3x + 6y + 2z &= 1 \\ x - y + 3z &= 2 \end{aligned}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Using simplex method, find optimal solution of $Z = 7x1 + 5x2$ Subject to $x1 + 2x2 \leq 6$ $4x1 + 3x2 \leq 6$ $x1 \geq 0, x2 \geq 0$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What do you mean by stationary points and inflection points? Using derivatives, find two numbers whose sum is 20 and sum of whose squares is minimum.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Using Simpson's $\frac{1}{3}$ rule evaluate $\int_0^1 \frac{1}{1+x} {dx}$ with 3 points of intervals. Find error of approximation. How many points are to be considered to make the approximation value within 10-5?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
